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WHEN AI MEETS MARKET REALITY

A 296K-Parameter CNN That Lost to a Coin Flip

EXECUTIVE SUMMARY

CONTENDER	CHALLENGER	RESULT
Deep CNN with GAF Imaging	**Random Walk**	**FAIL**
296,000 Parameters	Neutral Coin Flip (0.5)	Simplicity outperformed Complexity on every metric.

THE ARENA: VTS TOURNAMENT CONSTRAINTS

OBJECTIVE

Predict daily Bitcoin allocation weights (2016–2025). The goal is not just prediction, but profitable capital allocation.

THE METRIC

Recency-Weighted (RW) SPD Percentile.

Target > 60th Percentile

IRONCLAD RULES

1. Sum of Weights = 1.0
2. No Lookahead Bias (Strict Causality)
3. Long-Only Constraints

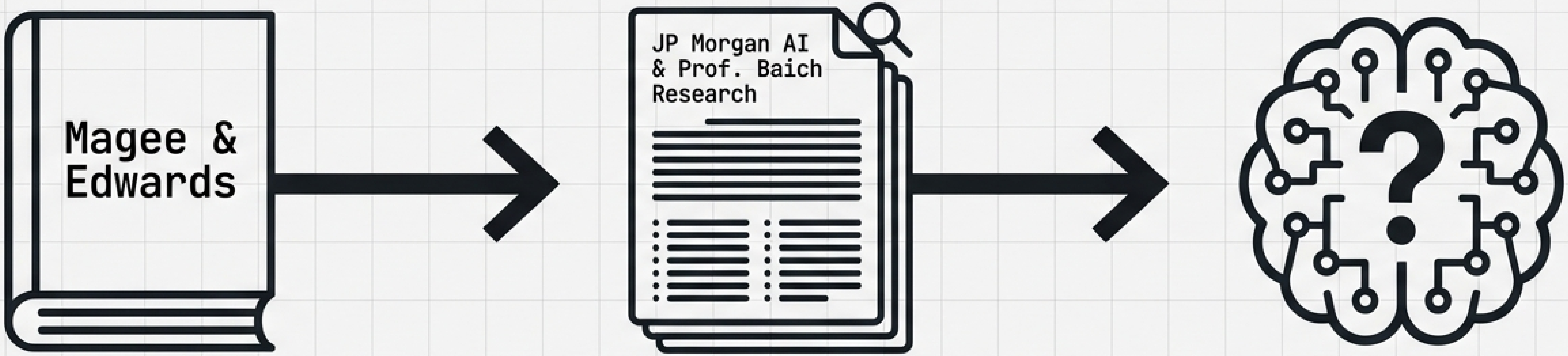
VERIFICATION

Leakage Test: PASSED

`Last-row modification difference: 0.00e+00`

THE HYPOTHESIS: CAN AI ‘SEE’ MARKET TRENDS?

THE FOUNDATION | THE ACADEMIC DISCOVERY | THE LOGIC GAP



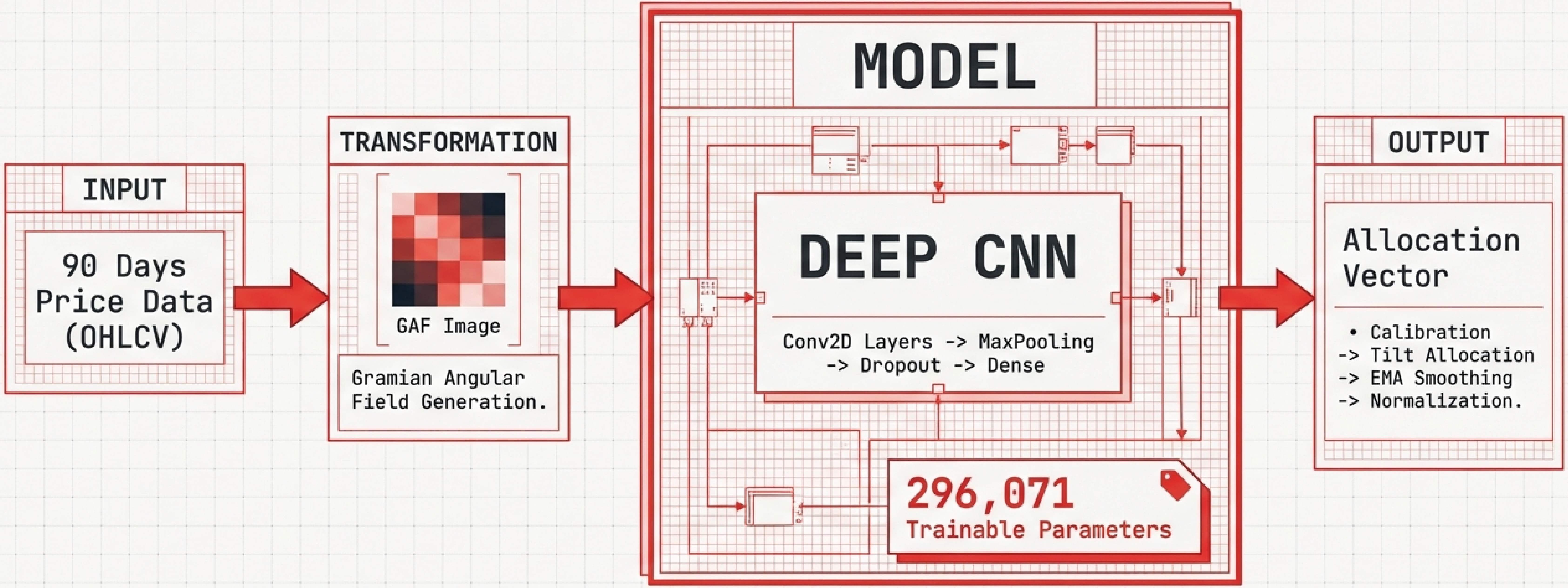
The Foundation: Human traders successfully ‘see’ visual patterns (Head & Shoulders, Support Levels) to make decisions.

The Academic Discovery: CNNs achieve 0.90+ F1 Scores on pattern classification.

The Logic Gap: Can classification success translate to profitable capital allocation under strict causality?

The premise: If humans use eyes, machines should use Convolutional Neural Networks.

THE ARCHITECTURE: 296,000 PARAMETERS



REALITY CHECK: THE MODEL FAILS TO LAUNCH

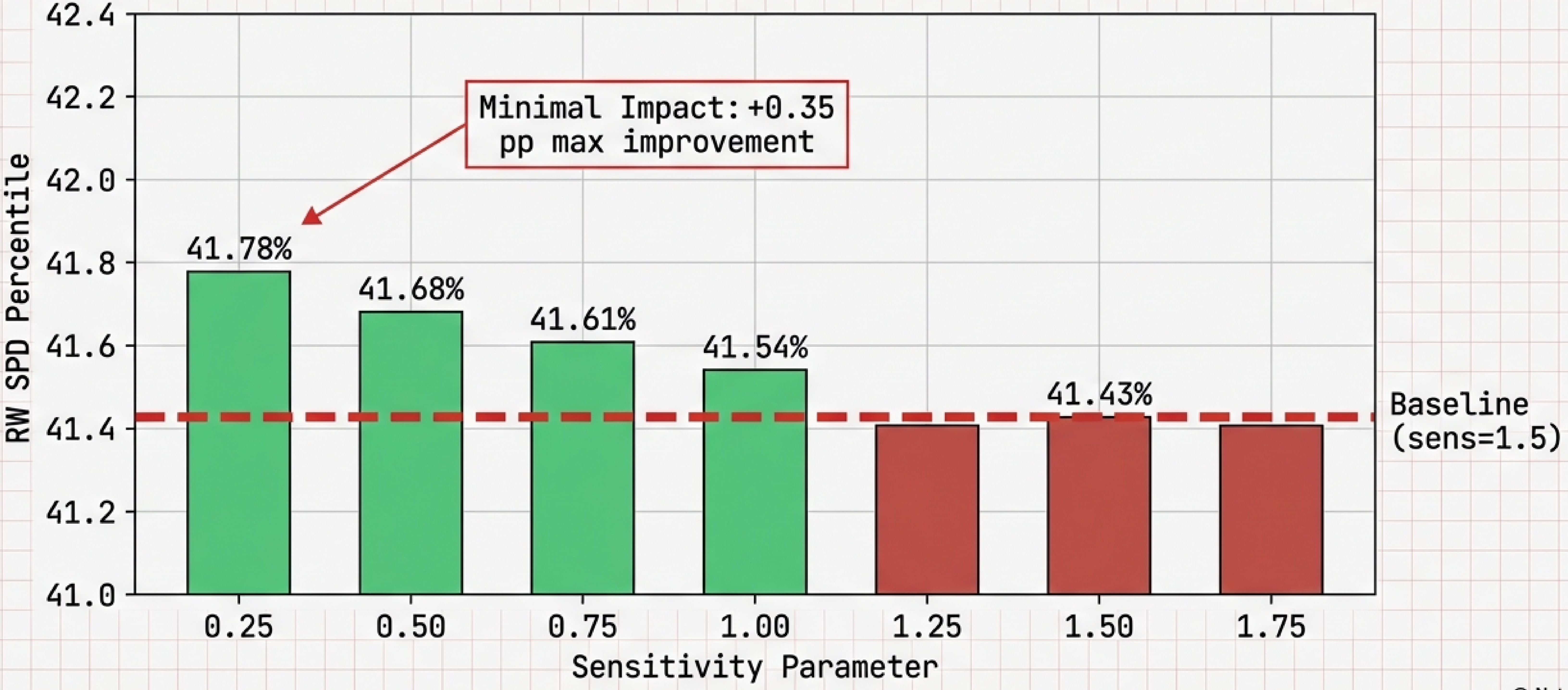
METRIC	CNN RESULT	TARGET	STATUS
RW SPD Percentile	41.43%	> 60.00%	FAIL
Win Rate vs DCA	54.32%	> 50.00%	MARGINAL
Windows Evaluated	3,076	N/A	COMPLETE

VERDICT: The sophisticated CNN underperforms simple Dollar-Cost Averaging.

INITIATING FORENSIC ABLATION STUDIES...

ABLATION STUDY #1: ALLOCATOR SIZING

Hypothesis: Is the betting strategy too aggressive?



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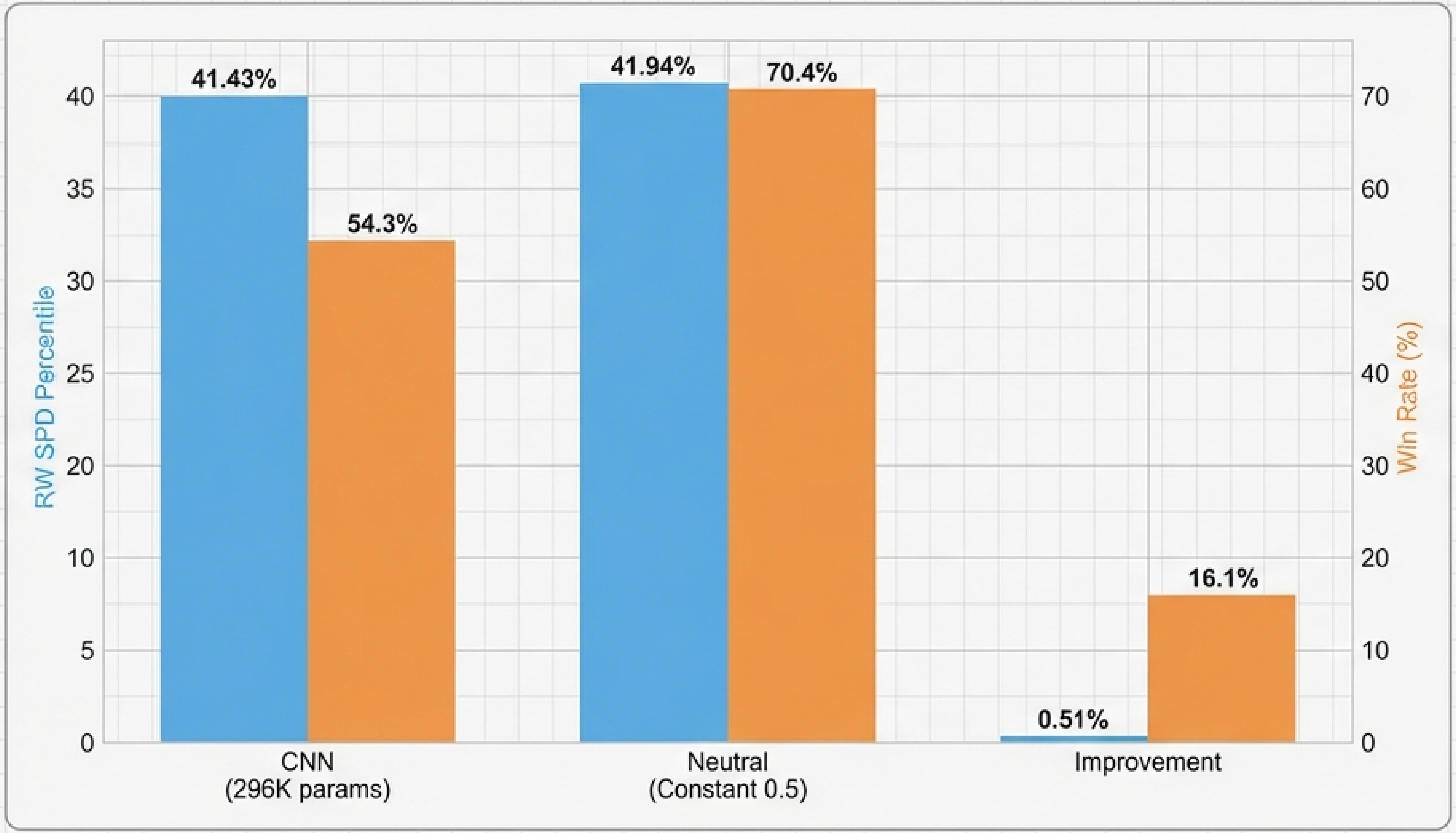
ABLATION STUDY #2: THE 'AHA!' MOMENT

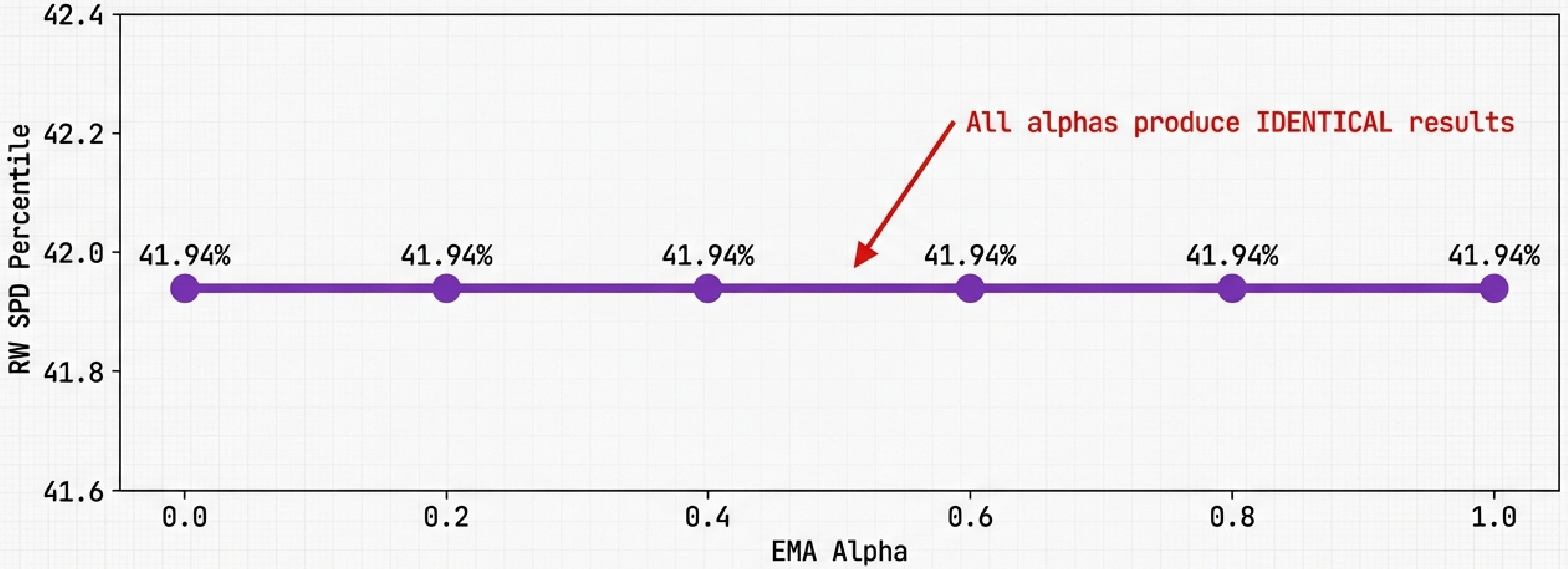
Hypothesis: Is the signal better than random guessing?



INSIGHT:
The Neutral model (Coin Flip) performs BETTER. The CNN signal is effectively noise.

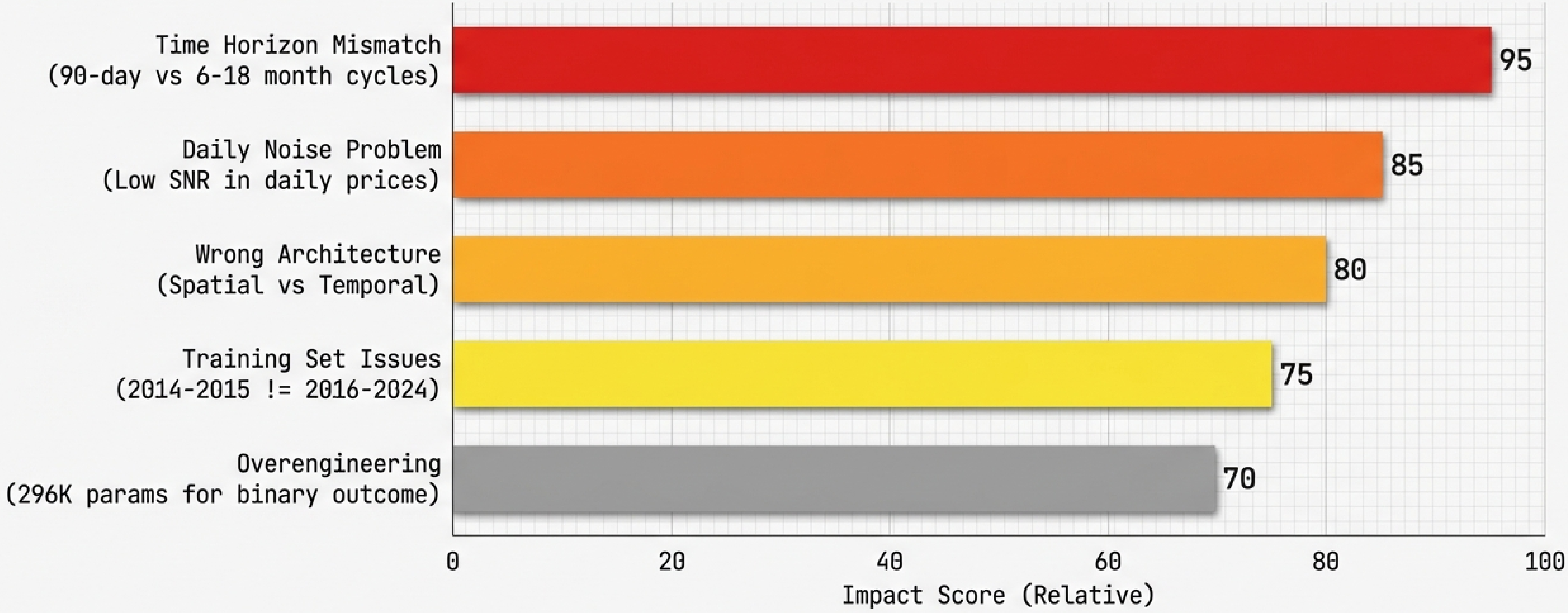
ABLATION STUDY #3: MATHEMATICAL PROOF OF FAILURE

Hypothesis: Can smoothing save the signal?



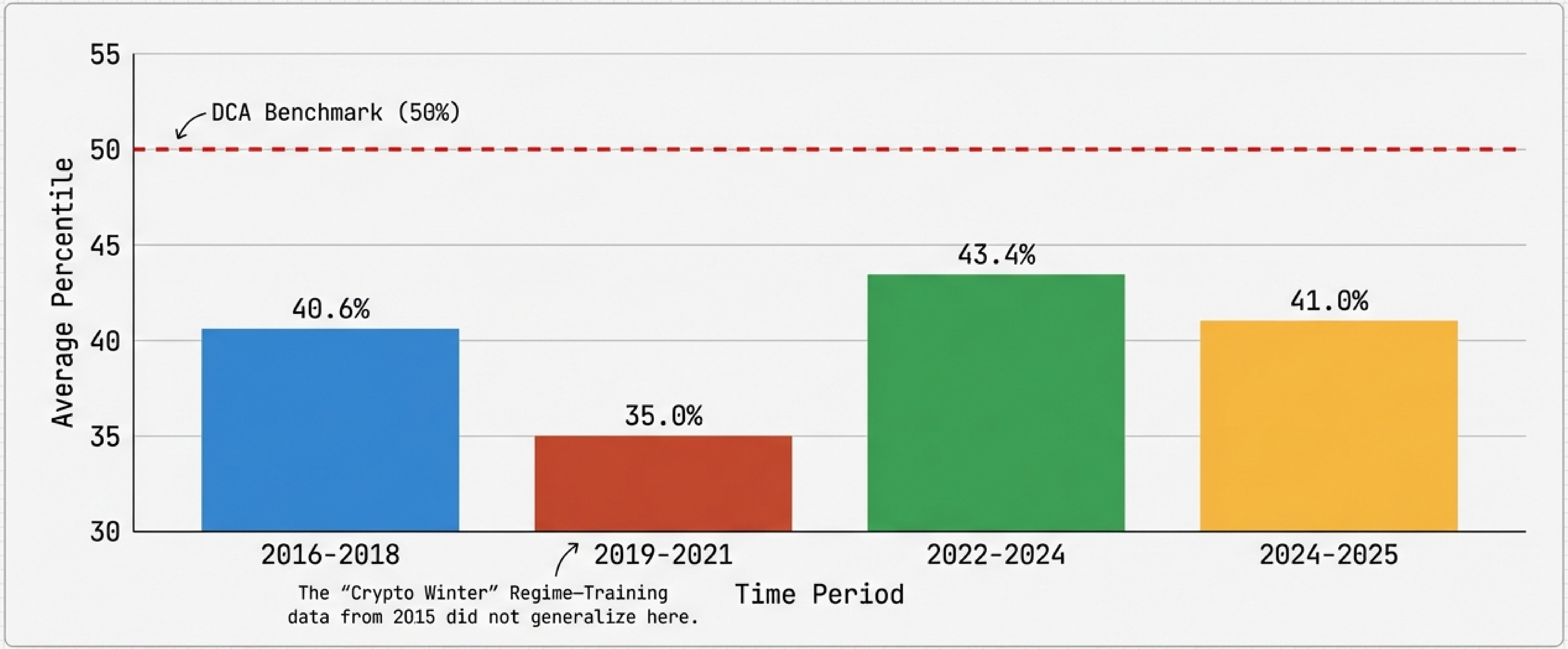
PROOF: If Signal is Random -> Tilt is Zero -> Smoothing a Zero Vector = Zero Effect.

THE AUTOPSY: ROOT CAUSE ANALYSIS



TEMPORAL ANALYSIS: U-SHAPED PERFORMANCE

Why the past didn't predict the future.



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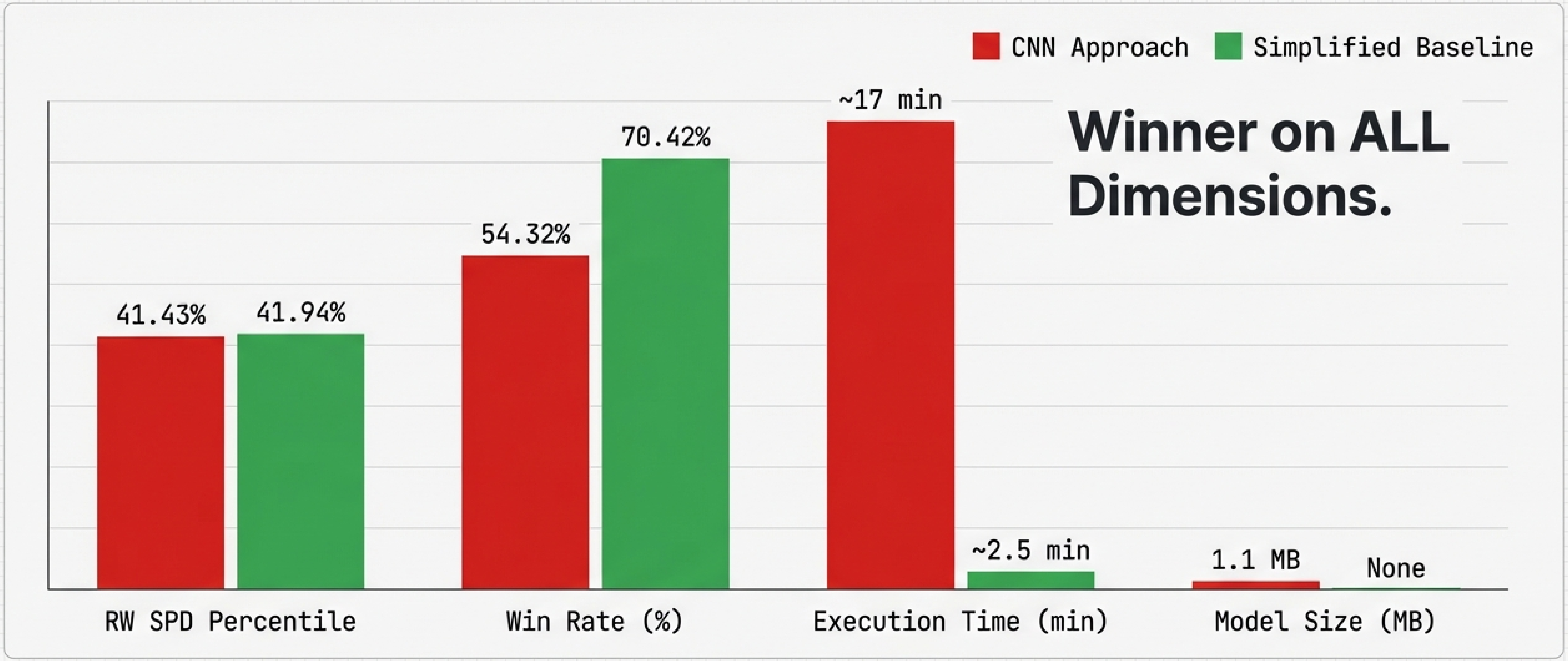
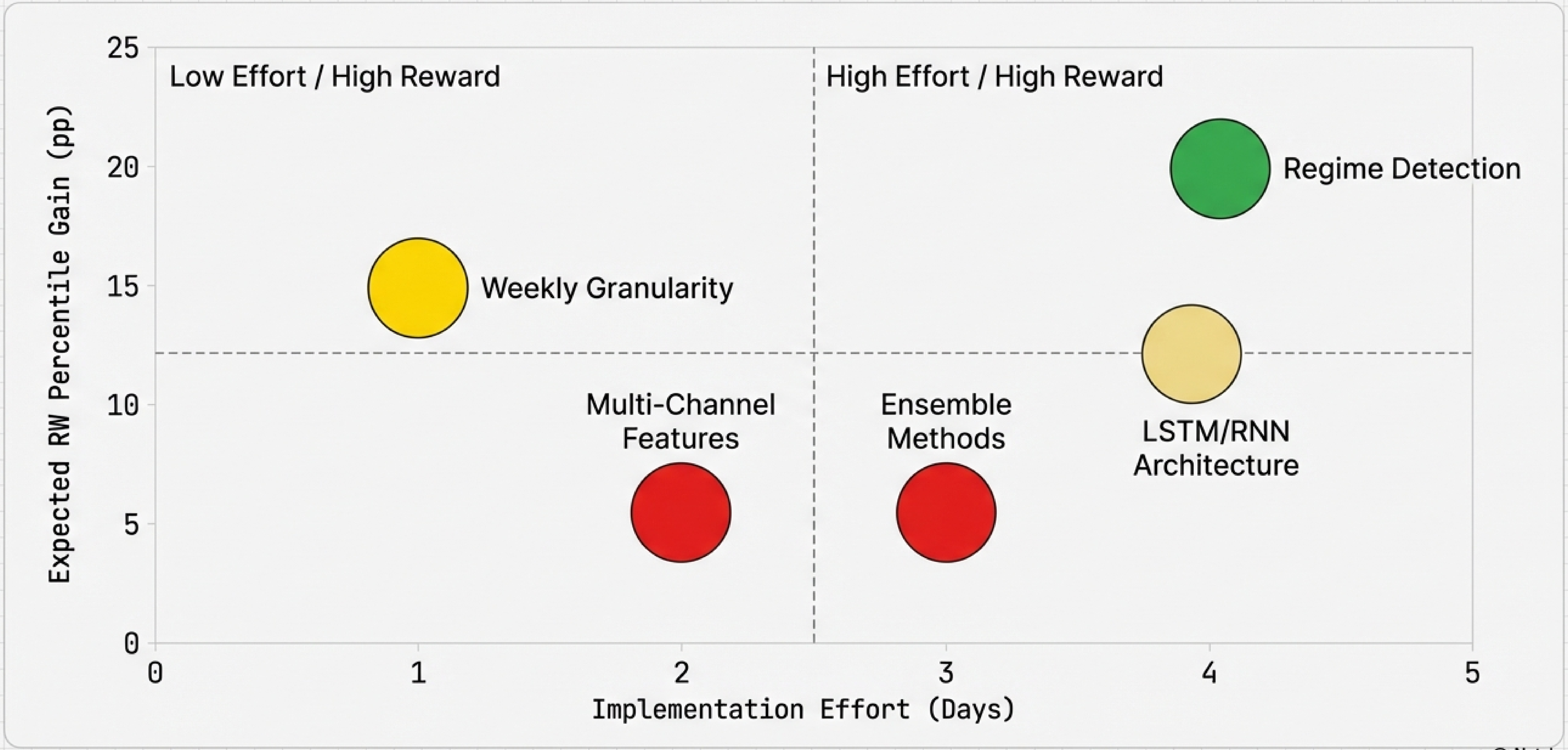
THE SOLUTION: SIMPLICITY WINS



FUTURE ROI: WHERE TO INVEST NEXT?



LESSONS FROM THE LAB

01 ABLATION STUDIES ARE CRITICAL

Without them, we would have tweaked parameters endlessly.
The studies proved the core signal was the problem.

02 TEST 'BETTER THAN RANDOM' EARLY

This should be Ablation #0. If you can't beat a coin flip,
don't build a neural network.

03 COMPLEXITY ≠ PERFORMANCE

Occam's Razor applies to trading. A 296K parameter model
provides no value if the underlying hypothesis is flawed.

THE BOTTOM LINE

> **"The best model is the one that works, not the one that sounds impressive."**
>
> We built a technically sophisticated system and discovered through rigorous testing that it performs equivalently to a coin flip. This simplified version represents our evidence-based conclusion: in financial time series, signal quality matters more than model complexity.

Code: `Visual_Trading_System repository`
Doc: `LESSONS_LEARNED.md`